

RISHABH SHARMA

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EDUCATION

Northeastern University, Boston

Master of Science, Software Engineering Systems

May 2026

3.7 GPA

Relevant coursework: OOD, Data Science, Data Structures, Cloud Computing, iOS Development, Web Development, UI/UX Design

Visvesvaraya Technological University, Bangalore

Bachelor of Engineering, Information Science

Aug 2019

TECHNICAL SKILLS

Programming: JavaScript, TypeScript, HTML, CSS, Java, Python, MySQL, MongoDB, PostgreSQL, Redis, C++, Swift, SwiftUI

Framework: React.js, RSC, Redux, MaterialUI, Node.js, GraphQL, Spring Boot, Spring MVC, OAuth 2.0, JUnit, Mockito, Kafka, Firebase, LangChain, RAG, Vector Search, LLM, REST API

Tools: GitHub, Git, Figma, Postman, AWS, DigitalOcean, Terraform, Packer, Lambda, CloudWatch, BigQuery, MCP Server, Chrome Extension (MV3), Gemini API, OpenRouter, Docker, NGINX, GitHub Actions, Gradle, OpenAPI 3.0, Playwright, Selenium

PROFESSIONAL EXPERIENCE

Mabl, Boston: Software Engineer Co-op

May 2025 - Dec 2025

- Built bidirectional Jira integration pulling test execution data from Google BigQuery to auto-generate detailed failure tickets with DOM screenshots, failure analysis, and run metadata, reducing manual ticket creation by 75% while maintaining real-time sync across platforms.
- Led end-to-end delivery of Xray test management integration, starting with customer discovery sessions to understand existing QA workflows, through feature flag-gated rollout controlling which Xray capabilities were exposed per customer, to production deployment and post-launch support resolving customer queries, reducing testing workflow overhead by 85%.
- Integrated Jenkins CI/CD pipeline with mabl test agent to automate test execution and error reporting, streamlining continuous testing workflows and enabling real-time visibility into test failures across development cycles.
- Navigated a large-scale production codebase to scope, implement, and ship datatable search and filtering enhancements, debugging cross-system issues in production and iterating on user feedback post-deployment, improving overall test authoring efficiency by 40%.

Publicis Sapient, Bangalore, India: Software Engineer

Jun 2021 - May 2023

- Owned full-stack delivery of customer-facing features end-to-end, designing concurrent Java Spring Boot microservices with MySQL/SQL Server while building responsive React.js and Redux components, improving interface load times by 20% and system scalability by 15%.
- Integrated UnionPay payment gateway into the microservices backend with a React frontend payment option, reducing transaction processing time by 25-35% and expanding payment coverage for international customers; automated deployment via Jenkins CI/CD, cutting release cycles from weeks to days.
- Integrated Elasticsearch for efficient full-text capabilities, improving search relevance by 25% and query performance by 40%.

Oracle Cerner, Bangalore, India: Software Engineer

Sep 2019 - Aug 2020

- Engineered "Sentinel" data channel security system safeguarding sensitive patient health records in a regulated healthcare environment, enhancing overall data security measures by 25%.
- Leveraged Hadoop MapReduce framework to orchestrate distributed data processing across multiple nodes, achieving 40% reduction in data clustering latency.
- Engineered fault-tolerant event-driven messaging pipelines using Apache Kafka across distributed nodes, achieving 99.99% reliability and decoupling critical service dependencies in a microservices architecture.
- Implemented load balancing algorithms, resulting in 35% increase in scalability.

ACADEMIC PROJECTS

TabSense: AI-Powered Semantic Tab Search

Feb 2026 - Apr 2026

- Built and shipped an AI-powered Chrome Extension (Manifest V3) with a Spring Boot GraphQL backend for semantic tab search, cutting information lookup from 3-5 minutes to 12-25 seconds per query (~85-93% faster).
- Engineered an end-to-end RAG pipeline (Playwright scraping, chunking, Gemini embeddings, cosine retrieval, OpenRouter generation), processing 5-20 tabs per sync and indexing 100-400+ chunks per run for citation-backed answers.
- Implemented Redis cache-aside with TTL and sync-version invalidation, reducing repeat-query latency from 6-9s to 0.6-1.5s (~75-90% faster) and cutting redundant LLM generation calls by 50-80% across follow-up queries.
- Hardened Gemini embedding calls with retry/backoff and pacing controls, dropping throttling-related failures from 30-45% to under 15% and lifting sync success to 85-95%; achieved P50 ~2-3.5s and P95 ~6-9s search latency.
- Improved extension UX with selective sync scope (active tab vs. full window), persistent popup state, and a tabId → URL → new-tab citation fallback, reducing repeated rework by ~40-60% in multi-tab sessions.

Job Finder Web Scraper

Nov 2025 - Jan 2026

- Architected a multi-service pipeline decoupling a TypeScript MCP server from a Java Spring Boot backend via a spec-first REST API with OpenAPI 3.0 and Swagger UI, implementing a prompt-driven orchestration layer enabling CLI natural language queries to trigger downstream Playwright workflows with contract-validated responses.
- Engineered a resilient DOM extraction layer using Playwright for dynamic JavaScript-rendered content with retry logic, request throttling, and selector fallback chains, applying the same automation patterns used in end-to-end UI testing with Selenium.
- Built an event-driven sync pipeline to Notion API persisting structured job records with deduplication logic, preventing redundant entries across concurrent scraping runs.

NimbusForge

Sept 2024 - Dec 2024

- Architected multi-AZ AWS infrastructure with Terraform provisioning VPC, ALB, Auto Scaling Groups (2-20 instances), and PostgreSQL RDS with automated failover, achieving 99.9% uptime resilient to single AZ failure.
- Engineered a zero-touch provisioning pipeline using Packer to bake immutable AMIs, eliminating environment drift across dev, staging, and production and cutting instance bootstrap time by 70%.
- Built a production-hardened Spring Boot REST API with Gradle build automation, role-based access control, BCrypt authentication, and AWS Secrets Manager for credential rotation, achieving sub-200ms p99 latency under load.
- Designed a zero-downtime CI/CD pipeline via GitHub Actions with canary instance refresh, Lambda-triggered SNS alerts, and CloudWatch dashboards, cutting deployment time by 85%.